

An explainable framework for assisting the detection of AI-generated textual content

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ABSTRACT

The recent development of generative AI (GenAI) algorithms has allowed machines to create new content in a realistic way, driving the spread of AI-generated content (AIGC) on the Internet. However, generative AI models and AIGC have exacerbated several societal challenges such as security threats (e.g., misinformation), trust issues, ethical concerns, and intellectual property regulation, calling for effective detection methods and a better understanding of AI-generated vs. human-written content. In this paper, we focus on AI-generated texts produced by large language models (LLMs) and extend prior detection methods by proposing a novel framework that combines semantic information and linguistic features. Based on potential semantic and linguistic differences in AI vs. human writing, we design our Semantic-Linguistic-Detector (SemLinDetector) framework by integrating a transformer-based semantic encoder and a linguistic encoder with parallel linguistic representations. By comparing a series of benchmark models on datasets collected from various LLMs and human writers in multiple domains, our experiments show that the proposed detection framework outperforms other benchmarks in a consistent and robust manner. Moreover, our model interpretability analysis showcases our framework's potential to help understand the reasoning behind prediction outcomes and identify patterns of differences in AI-generated and human-written content. Our research adds to the growing space of GenAI by proposing an effective and responsible detection system to address the risks and challenges of GenAI, offering implications for researchers and practitioners to better understand and regulate AIGC.

1. Introduction

AI-generated texts have proliferated on the Internet in recent years, and this trend has accelerated with the rapid development of generative artificial intelligence (AI) algorithms. For instance, OpenAI released GPT-3 in 2020, which contains 175 billion parameters based on Generative Pretrained Transformer (GPT), exceeding the number of neurons in a human's brain. With several input sentences serving as the "prompt," which instructs the model about what to do and triggers the subsequent output, GPT-3 can generate coherent and logically consistent content that can be challenging for humans to detect [1]. At the end of 2022, OpenAI released a more powerful model, ChatGPT, which was a revolution in generative AI (henceforth, GenAI) that brought AI-Generated Content (AIGC) into the limelight for the masses. The content generated by ChatGPT is significantly better than GPT-3, and in many cases, it is difficult for humans to confidently detect if AIGC is written by ChatGPT or a real person.

Since its emergence, GenAI, particularly large language models

(LLMs) for text generation, has received great attention from researchers and practitioners. The potential of GenAI has been examined across various task contexts, such as information retrieval, creative work, and even scientific research [2–4]. However, the fact that people cannot easily distinguish whether the content is AI- or human-authored has led to a general concern around severe social risks that may result from the abuse of GenAI. For example, Wang et al. [5] discussed key problems of GenAI, such as security threats (e.g., misinformation), trust issues due to bias and hallucination, ethical issues (e.g., plagiarism and phishing), and intellectual property regulation. Moreover, the proliferation of AIGC can make it easier to create and spread misinformation, leading to new types of criminal activities such as AI fraud and defamation.

These challenges caused by GenAI have attracted interest from both researchers and practitioners in the development of detection systems aimed at identifying texts generated by AI. These detection methods can be broadly classified into five categories based on the types of information relied on, including semantic features, linguistic features, LLM intrinsic features, watermarks, and combined features. Collectively,

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these methods offer valuable ways of identifying AI-generated texts; however, some notable gaps exist, which require a more systematic approach. First, most existing efforts in detecting AI-generated texts largely consider only one type of information (semantic information, linguistic features, or LLM intrinsic features) in isolation. While each aspect may uniquely contribute to the recognition of AI-generated texts, combining multiple types of information could further improve detection performance and robustness. In addition, although some studies leverage combined features [6,7], their methods do not offer end-to-end solutions and require the provision of a support corpus including both human- and AI-written texts in advance. As a result, the selection of support corpus (e.g., topics and number of documents) can significantly affect the performance of their detectors, which makes their method insufficient in many real-world use cases. Lastly, prior detection methods typically lack interpretability and transparency [8]. An interpretable detection framework with explicit decision rationales not only fosters trustworthy and responsible decision-making but also extends our understanding of what characterizes AI-generated content [9]. Thus, our research objectives are to: (1) propose a framework that utilizes both semantics and linguistic features to detect AI-generated texts without supplying a support corpus; and (2) use this detection framework to obtain interpretable predictions and identify the distinguishing patterns of AI-generated content.

Based on the semantic and linguistic differences between AI-generated vs. human-written content, we propose Semantic-Linguistic Detector (SemLinDetector), which fuses semantic and linguistic features. Specifically, we employed a pretrained language model, DeBERTaV3 [10], to capture semantic information. To comprehensively represent the possible linguistic features, we extract 16 linguistic representations in parallel via Textagon [11] and encode the rich linguistic features through Hierarchical Attention Networks (HANs). By fusing these two neural network components, we argue that the combination of semantic and linguistic features could potentially enhance detection performance by taking advantage of their complementary strengths.

Our experiments show that our proposed *SemLinDetector* achieves high detection performance that is robust when detecting the texts from different LLMs. More importantly, our proposed method consistently outperforms other benchmark models on out-of-sample testing cases, including student essays, news articles, and health conversations. We further employed Layer Integrated Gradients (LIG) [12,13] in our detection framework, demonstrating the model's interpretability in identifying salient patterns in AI-generated texts in contrast to human-written texts. The insights from our model could subsequently help drive more sophisticated detection models and facilitate our underlying understanding of AI text generation processes [14].

Our research adds to recent Information Systems (IS) research on GenAI by developing an effective and interpretable detection framework for AIGC. This framework supports the decision process behind AIGC detection, serving as a decision support system to address the challenges of GenAI and mitigate potential risks. Theoretically, our paper shows there are both semantic and linguistic differences between AI-generated and human-written texts. We expand upon such differences in the model interpretability analysis, which further identifies key patterns underlying AI-generated texts and extends our understanding of AIGC. We also offer practical implications for practitioners in detecting AIGC and regulating GenAI on social media and digital platforms.

2. Related literature

2.1. Generative AI and its influence

Generative AI (GenAI) refers to artificial intelligence that can create novel content such as images, texts and audio. It differs from other types of AI generally for pattern recognition, classification, and recommendations [15]. The history of GenAI can be traced back to the Hidden Markov Model (HMM), which generates sequential information such as

acoustic and time series data [16]. Later on, the development of GenAI has been significantly advanced by deep learning methods including the Variational Autoencoder (VAE), Recurrent Neural Network (RNN), Generative Adversarial Network (GAN), and Transformers [16,17]. In recent years, the emergence of Large Language Models (LLMs) [18] and Diffusion Models [19] created new milestones in this space, enabling GenAI models to produce realistic content difficult for even humans to recognize. Regarding this, Antaki et al. [20] designed an experiment to evaluate ChatGPT's performance in the Ophthalmic Knowledge Assessment Program (OKAP) examination, finding that ChatGPT only performed roughly 15 % worse than the average scores of humans.

The rapid development of GenAI has been shown to benefit society in many areas, including education, creative work, and business. Specifically, Lo [21] reviewed literature focusing on ChatGPT's impact on education and found benefits when it serves as an assistant for instructors and as a virtual tutor for students. In addition, George & George [22] focused on ChatGPT's benefits in several business sectors, such as e-commerce, entertainment, finance, health, and news. For example, the high-quality automated customer service solutions based on ChatGPT help businesses increase efficiency and decrease costs. Furthermore, Xu et al. [4] found that ChatGPT can reduce the information retrieval cost and achieve similar overall task performance compared with Google Search. Despite the popularity and power of GenAI, the rapid development of such a new technology has caused a series of public concerns related to the risks and consequences of using GenAI, such as misinformation, ethical concerns, and hallucinations [5,22,23–25].

2.2. AIGC detection methods

Due to the risks and consequences of GenAI, researchers have been evaluating potential detection methods to effectively differentiate between AI and humans, especially in the case of texts generated via LLMs. In this section, we review existing detection methods and classify them into five categories based on the information utilized, as shown in Table 1.

In the category of *Semantic Feature*, a large body of prior methods distinguish AI- and human-written content involving text [24–31]. For example, Guo et al. [24] fine-tuned the DeBERTaV3 [10], a pre-trained language model, to directly classify given sentences into AI-written and human-written classes. Zhu et al. [27] asked LLM to paraphrase sentences and decide if the sentences were written by AI based on whether the semantic similarity between the paraphrased version and the original version was higher than a threshold. By contrast, under the category of *Linguistic Feature*, researchers used structural language characteristics, such as syntax and grammar, to detect AI-written texts [32,33]. Fröhling and Zubiaga [32] used features including repetitiveness, syntactic and lexical diversity, coherence, and level of purpose in the input sentences to train a simple neural network model as their detector. Liu et al. [33] encoded coherence information in the form of a graph into a text representation to exploit the linguistic features and utilized such a text representation of input sentences to delineate AI versus human writing.

In the category of *LLM Intrinsic Features*, prior studies focused on statistical features such as the log-likelihood of the input sentences generated by a certain LLM [35,34]. For example, Hans et al. [35] proposed a statistical feature called Binoculars based on contrasting two closely related LLMs to separate the human- and AI-written texts. In comparison to utilizing intrinsic features, researchers in the category of *Watermark* proposed a method of inserting a “watermark” when LLMs generate texts that did not reduce the quality of the generated content [37]. The main idea was to use certain tokens when the LLMs generate texts and then conduct a statistical test to decide if the texts are generated by AI in the inference stage.

Furthermore, in the emerging category of *Combined Features*, researchers combined different features (semantic, linguistic, or LLM intrinsic features) for detection [6,7]. For example, Soto et al. [7]

Table 1
Summary of AI-generated texts detection methods.

Category	Paper	Method
Semantic Feature	Guo et al. [24]	Fine-tunes pre-trained language models such as RoBERTa to detect AI-generated texts.
Semantic Feature	Hu et al. [25]	Adopts an adversarial learning strategy to develop a paraphraser vs. detector framework. The paraphraser model deceives the detector, and the detector model classifies texts into three classes: human-written, AI-generated and paraphrased AI-generated.
Semantic Feature	Yang et al. [26]	Fine-tunes RoBERTa on their proposed HPPT dataset, containing human-written and ChatGPT-polished texts.
Semantic Feature	Zhu et al. [27]	Prompts ChatGPT to revise original texts and compares the similarity between revised texts and original texts. The original texts will be predicted as AI-generated if the similarity is higher than a threshold.
Semantic Feature	Yang et al. [28]	Given an original text, truncates it in the middle and then uses the preceding portion to prompt the LLM to generate another version of the text. Then conducts n-gram analysis between the generated version and the rest of the truncated original text to detect if the original text is generated by AI.
Semantic Feature	Mao et al. [29]	Prompts ChatGPT to revise an original text and compares the distance between the revised text and original text. Trains a Logistic Regression model based on the distance to detect.
Semantic Feature	Bhattacharjee & Liu [30]	Prompts ChatGPT to directly detect AI-generated texts.
Semantic Feature	Nguyen-Son et al. [31]	Measures the similarity between the input sentence and its generated counterpart by LLMs to detect analogous machine-generated sentences that closely mimic human-written ones.
Linguistic Feature	Fröhling & Zubiaga [32]	Demonstrates the differences between AI-generated and human-authored texts, such as repetitiveness, syntactic and lexical diversity, level of coherence, and level of purpose. Then trains ML classifiers based on linguistic features to perform detection.
Linguistic Feature	Liu et al. [33]	Constructs a coherence graph from a sentence and uses the coherence graph as input to train a ML model as the detector.
LLM Intrinsic Feature	Bao et al. [34]	Proposes a feature called conditional probability curvature to elucidate discrepancies in word choices between human-written and AI-generated texts.
LLM Intrinsic Feature	Hans et al. [35]	Proposes a feature called Binoculars derived from the perplexity and cross-perplexity from two LLMs.
Watermark	Kirchenbauer et al. [36]	Inserts a watermark when the LLM generates the texts. The watermark works by selecting a randomized set of “green” tokens before a word is generated, and then softly promotes the use of green tokens during sampling.
Combined Features	Soto et al. [7]	Learns the style representations of human-written texts and machine-generated texts following a contrastive learning strategy.
Combined Features	He et al. [6]	Develops ML classifiers based on the style representations to detect AI-generated texts.
Combined Features		Integrates retrieval-based techniques with other detectors, which enhances other detectors by fusing semantic information.

proposed a detection framework that first transforms a sample of AI and human written documents into both a semantic representation and linguistic representation (referred to as stylistic representation) in the support corpus. Then, for a new document during inference, the model transforms it into the semantic and linguistic representation and

computes the pairwise similarities between the inference sample representations and those in the support corpus. The new document is classified as AI-written if it is more similar to those AI-written documents in the support corpus.

Although prior research has developed various ways of detecting texts generated by LLMs, two notable gaps exist in prior detection methods. First, most of the existing detection methods only rely on one type of information, such as semantic information (words in sentences), linguistic features, or specific LLM intrinsic features, but rarely combine multiple categories of features together to augment detection performance. More importantly, the detection method relying on specific LLM features can be highly dependent on the LLM itself, and the detection model can be easily invalidated by new or evolved LLMs. Even though Soto et al. [7] combined semantic and linguistic features and He et al. [6] enhanced other detectors by inserting semantic information, both detection methods require the pre-selection and provision of both AI- and human-written documents as a support corpus in advance for comparison at the inference stage. As a result, these methods, by nature of their design, are subject to certain limitations. First, users of the detector must manually create a support corpus based on their task, which requires both domain expertise and additional effort, limiting ease of use. Second, the construction of a reliable ground truth is not feasible in many situations, particularly in settings like student essays where it is difficult to collect sufficient human- and AI-written samples in advance. Third, such a detector’s performance can be sensitive to the composition of the support corpus, which may lead to varying performance across domains or support corpus sizes. These challenges highlight the need for a more robust and generalizable end-to-end detection framework that combines multiple types of information.

Furthermore, existing methods typically lack interpretability, such that they only make predictions when given a document without articulating the rationale [8,38]. This limits a human’s ability to make responsible decisions based on the detection system while also making it difficult to learn the different patterns distinguishing AI-generated versus human-written content, thus inhibiting human understanding of the detection outcome [9,39]. To address these gaps, we propose a framework that combines both semantic and linguistic information to directly detect AI-generated texts and employs explainable AI algorithms to help users interpret the classification decision.

2.3. Semantic and linguistic differences in AI-generated texts

Our underlying rationale for combining semantic and linguistic information relies on the semantic and linguistic differences in AI-generated texts. Specifically, the underlying training and generation process of AI could potentially make it different from human in writing. To date, even though GenAI models have evolved over several generations with greater task performance, the creation of GenAI models in general and LLMs in particular remains similar and typically consists of two core components: pre-training and reinforcement learning with human feedback (RLHF) [40]. Although the process containing pre-training and RLHF has successfully created the state-of-the-art LLMs with strong capabilities across various tasks, this underlying process has several implications when compared with human writing processes.

Table 2
Examples of human and AI written texts (same prompt).

Prompt	
Discuss why having an archiving strategy for data is important for a business.	
Human Written	An archiving strategy for data is important because of the fact it would save all the important files needed for businesses, replacing file cabinets completely...
AI Written	Implementing a robust archiving strategy for data is a critical imperative for businesses in the digital age. Such a strategy involves the systematic, responsible, and organized management of a company’s digital assets...

Table 2 shows a pair of human-written and AI-generated texts given the same prompt to demonstrate the different patterns we discuss below.

First, the probabilistic process of sampling the next tokens given previous ones induces the model's tendency to choose mostly frequent words from the training data [41]. In contrast, human writers may determine their word choices based on familiarity, memory, or personal preference. Second, because of the statistical sampling process to choose subsequent tokens, LLMs can be more systematic in term use and sentence construction, limiting diversity but increasing complexity in language use. On the contrary, human writers can be arbitrary and flexible in some cases when considering terms and building sentences. Herbold et al. [42] found that ChatGPT-generated essays use more nominalizations and have higher sentence complexity, such as the use of more "scientific" language, whereas human-written essays make more use of modal and epistemic constructions. As shown in Table 2, the AI-written text uses a more complex sentence structure and more "scientific" language such as "robust" and "digital assets." Third, reinforcement learning with human feedback can instill human preference and instructions to limit the flexibility of model responses. In particular, the model can be instructed to generate more structured and formal content, restricting the use of extreme or emotional words to make it more accurate and useful. Comparatively, human writers may use more flexible structures and informal languages and adopt more extreme words with greater sentiment polarity. For example, Kabir et al. [43] found that AI-generated texts adopt more words related to analytical thinking and clout expressions, express more positive emotions, include more drive attributes that show the need, desire, and effort to achieve something, and use more formal words and expressions than human written texts. As shown in Table 2, the human-written text uses more informal language like "because of the fact" and "replacing the file cabinets completely."

These findings suggest that significant semantic and linguistic differences exist between AI-generated and human-written texts. These differences not only are useful sources to detect AI-generated texts, but they also improve our understanding of the nature of AIGC and our confidence in accurately recognizing AIGC. Thus, our proposed framework seeks to effectively combine both aspects. First, our framework leverages both semantic and linguistic differences for the detection of AI-generated texts and systematically rather than manually extracts features to discern the differences. Second, by designing the framework with clear roles for each component, we allow the framework to be explainable to generate insights about the differences between human and AI used for detection. In this regard, our framework builds upon prior findings and integrates the semantic and linguistic differences to both better detect and understand AI-generated texts.

3. Detection framework

In this section, we present the proposed AI-generated text detection framework, Semantic-Linguistic-Detector (SemLinDetector), in detail. Our detection framework combines semantic features and linguistic features into a unified architecture. Because linguistic features are represented at a higher level than semantic features, they require additional representation before they are fed into the neural-network-based framework. Therefore, we first introduce the linguistic parallel representation, which serves as a critical part of ensuring the framework's capability to extract linguistic features. Then, we discuss the architecture of the framework, including its semantic encoder, linguistic encoder, concatenation module, and classifier module.

3.1. Textagon parallel representations

To extract the linguistic features from the original texts, we followed Textagon [11] approach and represented the original texts in parallel based on a total of 16 different linguistic representation strategies. This allows us to incorporate linguistic features more comprehensively into

our detection framework. Such an approach can utilize more complex underlying linguistic features and better learn the relationships among these features compared with a manual selection process of linguistic features. For example, the parallel representation method can learn the underlying diversity of part-of-speech in the texts without manually computing the same linguistic feature as in Fröhling & Zubiaga [32]. The definition and computation process of each linguistic representation are discussed as follows (more details about the Textagon parallel linguistic representation can be found in [11]).

3.1.1. POS

The Part-of-Speech (POS) representations replace each word in the original text with its corresponding part of the speech (e.g., noun, verb, ...).

3.1.2. NER

The Named Entity Recognition (NER) representations replace each word in the original text with its entity type, such as location (LOC) and organization (ORG). If the word's entity type is out of the scope of the NER entity list, it is replaced with <NONE>.

3.1.3. WN-Affect

WN-Affect [44] represents the emotion of each word using WordNet resources. All the words in the original text are replaced with their corresponding emotions such as sadness, shame, and despair. The substitution <NONE> is used when no emotion is indicated.

3.1.4. Word & Sense

Word-sense disambiguation (WSD) determines the correct meaning of a word in context, which is crucial for both human understanding and various computational linguistic applications. If the word is not in the corpus, the original word is replaced with <NONE>.

3.1.5. Hypernym

In the Hypernym representation, WordNet [45] is used to substitute the original word with its hypernym; for instance, the word "cat" is substituted with its hypernym "animal." Words lacking a hypernym are tagged with <NONE> in place of the original word.

3.1.6. Sentiment

In the Sentiment representation, each word is tagged by the combination of high/mid/low positive and high/mid/low negative labels based on positive vs. negative VADER (Valence Aware Dictionary and sEntiment Reasoner) sentiment annotations [46]. Words without sentiment tags are tagged using <NONE>.

3.1.7. Lexicon

In the series of Lexicon – <lexicon name> representations, we build ten lexicons via different definitions and tag each word with its corresponding lexicon name. For words excluded in the specific lexicon, we tag them with <NONE>. Lexicons include GLOVE [47], LIWC [48], and domain-specific examples such as ADR [49].¹

3.2. SemLinDetector architecture

To include both the semantic and linguistic information in the

¹ ADR refers to adverse drug reaction tags; AILEXCAT and AILEXINT refer to affect intensity lexical categorization and affect intensity lexical intensity respectively; GLOVECC, GLOVETW, and GLOVEWG are Common Crawl, Twitter, and Wikipedia plus Gigaword labels derived by clustering the tokens based on the GLOVE embedding; EMOLEX refers to NRC emotion lexicon; LIWC is the linguistic inquiry and word count categorization; SAVLEX refers to SAVLEX word standardization; SYN refers to synonym cluster label tags derived by clustering tokens based on their Synsets.

document for detection, we propose a fused model *SemLinDetector* based on the Decoding-enhanced BERT with Disentangled Attention (DeBERTaV3) and Hierarchical Attention Networks (HANs). Our proposed model consists of four major components: (1) Semantic Encoder, (2) Linguistic Encoder, (3) Concatenation Module, and (4) Classification Module. The Semantic Encoder and Linguistic Encoder are designed to independently capture the semantic and linguistic features of input texts, encoding them into separate vector representations. These two vectors are then fused in the Concatenation Module to form a unified representation that integrates both types of information. This fused vector from the Concatenation Module is subsequently transformed into a 2-dimensional output, where each dimension represents the probability of the input text being written by a human or AI, respectively. The following sections provide a detailed discussion of each component in the architecture, which is shown in Fig. 1.

3.2.1. Semantic encoder

For the encoding of the semantic information, we leverage the pre-trained language model, DeBERTaV3 [10], to encode the original document. Several existing semantic information-based detectors [24,50,51] have been built upon RoBERTa [52], and their experiments have demonstrated the outstanding ability to differentiate between AI-generated and human-written content. We adopt DeBERTaV3 instead of RoBERTa due to its superior performance in modeling token dependencies through disentangled attention mechanisms, enhancing the encoding of semantic nuances in the original document. In addition, this particular language model provides a good balance between the need to

effectively capture semantics and the size of the model for efficient training. As a pre-trained model on a large text corpus to learn rich semantic knowledge, it can encode the texts while simultaneously representing the underlying semantic information.

The DeBERTaV3 model follows the same architecture as BERT [53], consisting of multiple consecutive layers of *transformer blocks* [54], where the output of the previous transformer block is the input of the next one. For each transformer block, it first applies the multi-head attention operation to learn the underlying contextualized information from the output of the preceding transformer block. The first transformer block learns from the word embedding of the input texts.

We denote the output of the DeBERTaV3 model as $H_{deberta} \in \mathbb{R}^{l \times 768}$, where the l is the number of tokens after tokenizing the input texts, and 768 is the hidden size, which is a hyperparameter defined during the pre-train stage of DeBERTaV3. To make the semantic encoding compatible with the output of other modules in the architecture such as the linguistic encoder for the purpose of concatenating, we apply a pooling layer and a linear transformation to transform the 2-dimensional matrix representation $H_{deberta}$ into a 1-dimensional vector representation. The formula is defined as follows:

$$H_{sem} = \text{AveragePooling}(H_{12})W_{sem} + b_{sem} \quad (1)$$

$$\text{AveragePooling}(H_{deberta}) = \left[\frac{1}{k} \sum_{m=0}^{k-1} H_{deberta}[0m], \dots, \frac{1}{k} \sum_{m=0}^{k-1} H_{deberta}[l-1m] \right]^T \quad (2)$$

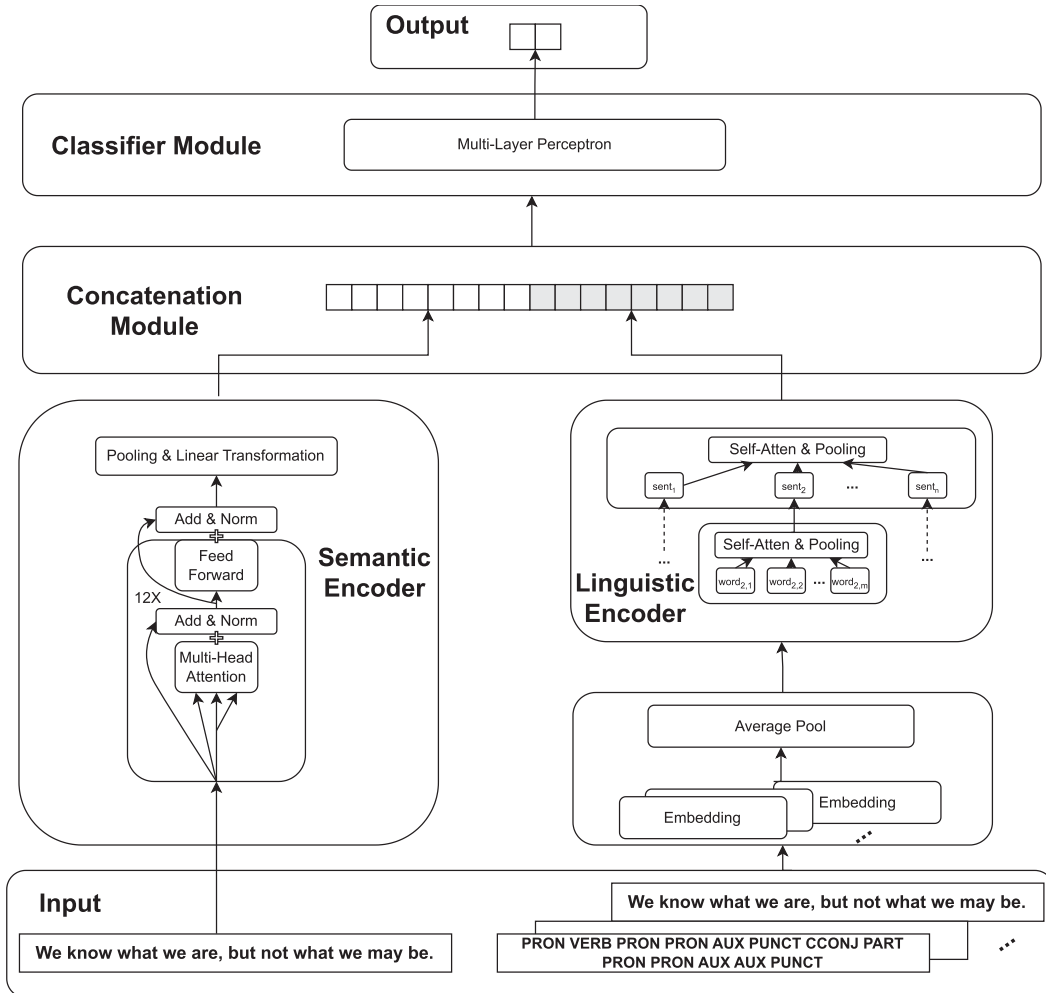


Fig. 1. The architecture of the SemLinDetector.

where the function *AveragePooling* reduces the input matrix's dimension from 2 to 1 by averaging the second dimension of the input matrix, k is the hidden size, which is 768 in DeBERTaV3, $W_{sem} \in \mathbb{R}^{768 \times 256}$ and $b_{sem} \in \mathbb{R}^{256}$ are the weights and bias respectively to transform the 768 dimension vector into 256 dimensions. $H_{sem} \in \mathbb{R}^{d \times 256}$ refers to the semantic encoding.

3.2.2. Linguistic encoder

The parallel representations derived from Textagon contain rich linguistic features and structure information at both word and sentence levels. To learn the linguistic and structure information, we design a neural network architecture based on Hierarchical Attention Network [55] by substituting the Gated Recurrent Network [56] with Self-Attention Network [54]. Such an adaptation can largely decrease the computational cost and execution time in the training process because the recurrent neural network is not suitable for parallel computing.

To transform the discrete linguistic representations into a derivable matrix, we adopt the idea of word embeddings. Specifically, for each linguistic representation, we first initialize a dictionary, where each key of the dictionary is a distinct linguistic label, and the value of a key is a randomly initialized vector which can be optimized during the training process. The embedding process for each linguistic representation can be formalized as:

$$E_{lin^i} = \text{Embedding}_i(\text{linguistic}_i) \quad (3)$$

where linguistic_i refers to the i -th linguistic representation of the input texts, $E_{lin^i} \in \mathbb{R}^{d_{embed}}$ and d_{embed} is the dimension of the word embedding which is set to 128. Then, we average the 16 parallel presentations' embeddings to get the unified embedding of all the Textagon linguistic features. The final linguistic embedding is:

$$E_{lin} = \frac{\sum_{i=1}^{16} E_{lin^i}}{16} \quad (4)$$

With the final embedding of the linguistic features, E_{lin} , we leverage the self-attention network to learn the underlying internal linguistic information for better encoding performance. It consists of two stages, focusing on the word-level and sentence-level information, respectively, within the input texts. In the first stage, which encodes the word-level information inside one sentence of the input texts, the formula is defined as follows:

$$A^j = \text{AveragePooling}\left(\text{SelfAttn}\left(E_{lin}^j\right)\right) \quad (5)$$

where $E_{lin}^j \in \mathbb{R}^{d_{le} \times d_{la}}$ refers to the averaged embedding of the j -th sentence for the focal input parallel linguistic representation. It is notable that the E_{lin}^j is part of the E_{lin} , which keeps only the embeddings of linguistic features in the sentence j . The $A^j \in \mathbb{R}^{d_{la}}$ refers to the output of the self-attention network representing the contextualized information in the j -th sentence of the parallel linguistic representation. d_{le} and d_{la} refer to the dimension of the linguistic tokens' embedding and the hidden state in the self-attention network, respectively. The self-attention network can dynamically weigh the different parts of the input matrix, enabling it to capture the contextual relationship between linguistic features at different positions in the sentence. We denote the self-attention operation as the function $\text{SelfAttn}(\cdot)$, which is defined as:

$$Q = W^Q X \quad (6)$$

$$K = W^K X \quad (7)$$

$$V = W^V X \quad (8)$$

$$\text{SelfAttn}(X) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} V\right) \quad (9)$$

where W^* refers to the parameters in the linear transformation to get query Q , key K , and value V . Similar to the pooling layer in the semantic encoder, the function $\text{AveragePooling}(\cdot)$ is employed to reduce the dimension for the ease of fusion with other modules, by averaging all words' hidden states in sentence j into one vector to represent the focal sentence.

In the second stage, the linguistic encoder learns the sentence-level information and the formula is defined as follows:

$$A = \text{Concat}(A^0, \dots, A^{s-1}) \quad (10)$$

$$H_{lin} = \text{AveragePooling}(\text{SelfAttn}(A)) \quad (11)$$

where $A \in \mathbb{R}^{d_{la} \times d_{la}}$ is the two-dimension document-level encoding of the parallel linguistic representation by simply concatenating all one-dimension sentence-level encodings A^j . Lastly, $H_{lin} \in \mathbb{R}^{d_{lin}}$ is the final encoding of the parallel linguistic representation.

3.2.3. Concatenation module

The concatenation module aims to fuse the hidden encodings of both the semantic information and the parallel linguistic representation. In this module, we directly concatenate the semantic encoding and the parallel linguistic representation encoding. Suppose that the semantic encoding is denoted as H_{sem} and the parallel linguistic representation encoding is denoted as H_{lin} . The formula to concatenate them is defined as follows:

$$H = [H_{sem}; H_{lin}] \quad (12)$$

where H is the aggregated encoding, including both semantic and linguistic information.

3.2.4. Classifier module

In the classifier module, we use a Multi-Layer Perceptron (MLP) as a classifier to distinguish between AI-generated and human-written content by treating it as a binary classification task. The MLP is effective in reducing the dimension of a given vector so that we can obtain a vector of logits representing the probability of each class. The aggregated encoding H from the concatenation module is used as the input in the MLP classifier, and the procedure is described by the following equation:

$$\text{Logits} = W_2(\text{ReLU}(W_1 H + b_1)) + b_2 \quad (13)$$

where W^* and the b^* are the parameters in the linear transformation, $\text{ReLU}(\cdot)$ is the activation function with rectified linear unit and $\text{Logits} \in \mathbb{R}^2$ is the vector of logits representing the probabilities of AI-generated and human-written content. In the end, we adopt the cross-entropy loss as the loss function to train the proposed detector.

4. Experiments

In this section, we present our experiments to evaluate the performance of the proposed *SemLinDetector*. We first introduce our data collection process involving several datasets used for training and evaluating the framework. We then describe our evaluation strategy, including the benchmark models and evaluation metrics, and finally discuss the experimental results.

4.1. Data

4.1.1. Training dataset

We created our dataset based on the same idea of a public AI-generated text detection dataset: Human ChatGPT Comparison Corpus

(HC3) [24], containing triplets of prompts, human answers, and AI answers. The creation of HC3 involved collecting question-answer pairs from Reddit and then using the questions as prompts to obtain AI answers. However, the ChatGPT model has been continuously improved since its release; as such, the ChatGPT-generated texts in the original HC3 dataset are now outdated. In our analysis, we used the ChatGPT-4o API to regenerate and update the ChatGPT-generated texts in the original HC3 dataset with a more recent model.

Notably, it is possible that the responses from the GenAI model are simply longer than those written by humans, making its output not comparable with human answers or driving text length to be an influential feature in detection. To avoid this problem, we limited the length of AI responses to less than 120 % of corresponding human texts in the API configuration. Furthermore, for the human-written answers, multiple answers were available for each question in the original HC3 dataset, and we randomly selected only one answer to construct matched comparison pairs, with each pair involving only one AI-generated answer and one human-written answer. As demonstrated in Table A1 in Appendix A, models trained on datasets with this restriction exhibit nearly the same performance on test datasets, regardless of whether a length restriction is applied. This supports the validity of the restriction and suggests it does not compromise model effectiveness.

Our final dataset includes 48,384 documents (24,192 pairs of AI and human answers), and we analyze the texts at the document level rather than splitting them into sentences to ensure the meaningfulness of linguistic features. The regenerated HC3 dataset was then partitioned into training (HC3-Train, 70 %), validation (HC3-Val, 10 %), and testing (HC3-Test, 20 %) subsets. Notably, each pair of AI and human answers to the same question resides only within the same subset (i.e., training, validation, or testing).

4.1.2. Evaluation dataset

As mentioned above, HC3-Test is used to evaluate the performance of our *SemLinDetector* framework. However, it was primarily applicable to assess the model performance within the same version of an LLM, as both training and testing datasets were collected. To evaluate detection performance on other variations of LLM, we followed the same strategy in constructing the HC3-Test dataset but replaced ChatGPT-4o with ChatGPT-4o-mini, Llama-3 (Llama-3.3-70B-Instruct) and Qwen-2.5 (Qwen-2.5-72B-Instruct), respectively, to create three additional datasets: HC3-Test-4o-mini, HC3-Test-Llama, and HC3-Test-Qwen.

The four test datasets above are *in-sample* datasets since they come from the same sample of HC3. To provide robust evaluations of our proposed *SemLinDetector* on *out-of-sample* datasets, we constructed four additional test datasets with different topics and genres. First, as part of a separate project that is under preparation by one of the authors on this article, we have access to data from a field study conducted in a real-world classroom setting involving primarily undergraduate students. These data are made up of essay responses to a set of 10 unique prompts drawn from an introductory-level business IT course (e.g., “Discuss why having an archiving strategy for data is important for a business”). This dataset includes 226 essay responses that students indicated were entirely human-written (i.e., with no AI assistance) and 127 essay responses that they indicated were entirely AI-written. Using these data from classroom setting, we created two out-of-sample datasets: (1) Students & Synthetic AI, which includes the 226 human-written essay responses matched with 226 synthetic AI-generated responses that all used ChatGPT-4o (452 in total); and (2) Students & Student Chosen AI, which includes the same 226 human-written essays as well as the 127 student-provided, AI-generated responses using a wider variety of AI-generated tools (353 in total).

In addition, we incorporated two other datasets from different domains at a larger scale: (1) CC News, and (2) Mental Health Conversation. Our CC News [57] dataset contains a subset of 10,000 randomly selected articles from the original CC News data (with a total of 708,241 English news articles between Jan 2017 and December 2019). The

dataset includes a sample of news articles with title, article text, and a summary of the news event. To construct the corresponding AI-generated news, we prompted GPT-4o with the news title and the summary. Finally, our CC News test dataset consists of 10,000 pairs of human-written and AI-written news items. For our Mental Health Conversation [58] dataset, it contains both human- and AI-written responses to mental health queries. The AI-written responses in this dataset were generated by a different LLM, Mistral-Instruct-v2-7B, through various prompts, suitable to serve as an additional out-of-sample test dataset. This dataset consists of 3507 pairs of human expert responses and AI-generated responses.

4.2. Training details

The proposed *SemLinDetector* was trained using the Hugging Face’s Trainer, and the optimizer was AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1e-8$) with the learning rate $2e-5$. The model was trained with 5 epochs with a batch size 16. All the training and experiments were conducted on an NVIDIA RTX 3090 GPU with PyTorch version 2.6.0 and CUDA version 12.6. We also set up the random seed as 42 for reproducing all the experimental results.

4.3. Benchmark models and evaluation strategy

To evaluate the performance of the proposed *SemLinDetector*, we choose a series of benchmark models that only rely on one type of information, i.e., linguistic features (LF) or semantic information (SI). Specifically, we chose three different models capitalizing on LF: (1) Logistic Regression, (2) XGBoost, and (3) Linguistic Encoder, which is one of the main modules of our *SemLinDetector* shown in Fig. 1. For Logistic Regression and XGBoost, we extracted over 220 handcrafted linguistic features through Linguistic Feature ToolKit (LFTK) [59]. We utilized features from LFTK rather than the parallel linguistic representations adopted in the proposed model because they are numerical linguistic features that align with the input for those statistical machine learning algorithms. In contrast, the third benchmark relying on LF, Linguistic Encoder, is one component in the *SemLinDetector*, as shown in Fig. 1, that uses the Textagon linguistic features as input. For SI, we selected two benchmark models: (1) Long Short-Term Memory (LSTM) and (2) Semantic Encoder in the *SemLinDetector* framework. Both models use the original texts as the input features. We also proposed another bi-modal benchmark model, Transformer + LFTK, which leverages both original texts and linguistic features extracted from LFTK to evaluate the effectiveness of a parallel linguistic representation compared with explicit linguistic features such as LFTK.

In addition to the benchmark models discussed in the paragraph above, we selected several detectors as benchmark models from peer-reviewed journals and conferences, including CoCo published in EMNLP 2023 [33], *Binoculars* published in ICML 2024 [35], and *PR-HPPT* published in APSIPA [26].² These models are also discussed in the literature review section. To evaluate the *SemLinDetector* and benchmark models, we used Macro-F1 and area under curve (AUC).

5. Results

Table 3 shows the Macro-F1 scores and AUC of our proposed *SemLinDetector* and other benchmark models on the in-sample datasets discussed in the above section, while Table 4 shows the performance on out-of-sample datasets. For the Logistic Regression, XGBoost, Linguistic

² For CoCo, we trained the model following their GitHub instructions (<https://github.com/YichenZW/Coh-MGT-Detection>). For *Binoculars* and *PR-HPPT*, we directly retrieved their models and performed inferences on our out-of-sample datasets (<https://github.com/ahans30/Binoculars> and <https://github.com/FreedomIntelligence/ChatGPT-Detection-PR-HPPT>, respectively).

Table 3

Detector performance across models on in-sample test datasets.

Model	HC3-Test		HC3-Test-4o-mini		HC3-Test-Llama		HC3-Test-Qwen	
	F1	AUC	F1	AUC	F1	AUC	F1	AUC
Logistic Regression	0.8337	0.9146	0.8494	0.9311	0.7981	0.8903	0.8174	0.9039
XGBoost	0.8695	0.9469	0.8807	0.9583	0.8411	0.9246	0.8578	0.9387
Linguistic Encoder	0.8988	0.9626	0.8895	0.9584	0.8639	0.9413	0.8756	0.9494
LSTM	0.9260	0.9778	0.9214	0.9778	0.8988	0.9630	0.9173	0.9761
Semantic Encoder	0.9815	0.9986	0.9925	0.9998	0.9862	0.9990	0.9917	0.9996
Transformer + LFTK	0.9830	0.9982	0.9916	0.9996	0.9858	0.9990	0.9907	0.9995
SemLinDetector	0.9879	0.9989	0.9941	0.9998	0.9823	0.9987	0.9927	0.9996

Table 4

Detector performance across models on out-of-sample test datasets.

Model	Students & Synthetic AI		Students & Student Chosen AI		CC News		Mental Health Conversation	
	F1	AUC	F1	AUC	F1	AUC	F1	AUC
Logistic Regression	0.7699	0.9717	0.6996	0.8764	0.4511	0.4066	0.6633	0.6765
XGBoost	0.8318	0.9930	0.7746	0.9229	0.5137	0.8607	0.7045	0.8183
Linguistic Encoder	0.8265	0.9515	0.7612	0.8899	0.7466	0.8439	0.7699	0.9061
LSTM	0.8289	0.8868	0.6966	0.7975	0.5498	0.7990	0.7560	0.9298
Semantic Encoder	0.9423	0.9883	0.9045	0.9660	0.8616	0.9884	0.9743	0.9969
Transformer + LFTK	0.9176	0.9769	0.8889	0.9606	0.8000	0.9553	0.9718	0.9970
CoCo [33]	0.9161	0.9079	0.6265	0.7038	0.6881	0.7605	0.9542	0.9429
Binoculars [35]	0.9579	0.9848	0.9247	0.9698	0.9579	0.9849	0.9618	0.9982
PR-HPPT [26]	0.8540	0.9464	0.8607	0.9542	0.6870	0.9707	0.9562	0.9979
SemLinDetector	0.9623	0.9956	0.9305	0.9744	0.9688	0.9955	0.9795	0.9978

Encoder, LSTM, Semantic Encoder, Transformer + LFTK, and SemLinDetector, we trained them on the HC3-Train dataset, and display their performance on all in-sample datasets as shown in Table 3. For the CoCo, Binocular, and PR-HPPT models, we only tested on out-of-sample datasets since they were not trained on the HC3-Train dataset, making it unfair to compare them on those in-sample datasets.

5.1.1. General observations

When considering performance within the same ChatGPT model (i.e., HC3-Test), our proposed *SemLinDetector* outperforms all other benchmark models and achieves excellent detection performance. This implies that the linguistic features provide useful information complementary to distinguishing AI-generated content from human-written content. When considering detection performance on samples from variations of LLMs (i.e., HC3-Test-4o-mini, HC3-Test-Llama, and HC3-Test-Qwen), our *SemLinDetector* outperforms all benchmark models except the Semantic Encoder (similar performance with very minimal difference). This proves our proposed model's ability to consistently identify the AI-generated content from different variations of LLM based on the training corpus from one LLM. Although the Semantic Encoder outperforms our *SemLinDetector* on certain in-sample datasets, its performance degrades significantly on out-of-sample datasets. Notably, the magnitude of performance reduction is much greater than the in-sample improvement, suggesting potential overfitting of the Semantic Encoder model. In addition, the performance of these detectors on all in-sample datasets are very close, which implies that different LLMs generate texts in a generally consistent manner, and the pattern derived from one LLM can be generalized for other LLMs. When considering detection performance on out-of-sample datasets, our *SemLinDetector* performs significantly better than most of the benchmark models, which implies that our model keeps high performance on samples with different topics and writing genres. Although one exception exists (*SemLinDetector*'s AUC is 0.9978, 0.0004 lower than Binocular), our proposed model demonstrates consistently high performance on the out-of-sample test datasets.

5.1.2. Ablation study

Compared the *SemLinDetector*'s sub-modules, namely the Semantic Encoder and the Linguistic Encoder, the experiment indicates that the integrated model, *SemLinDetector*, outperforms its individual components, which implies the validity of combining both semantic and linguistic features on the task of detecting AI-generated texts. It is also notable that incorporating the LFTK features with transformer (i.e., Transformer + LFTK) underperforms *SemLinDetector*, even though LFTK extracts over 220 linguistic features, which is much more than the 16 parallel representations from Textagon used in *SemLinDetector*. This is likely not only because simply feeding more numeric features is not always desirable in deep neural networks, consistent with [60], but also due to the more effective extraction and representation of linguistic features in Textagon. This observation validates the effectiveness of our design of parallel linguistic representation in *SemLinDetector* compared to using more human-crafted features in LFTK.

Overall, our proposed *SemLinDetector* framework achieves consistently better performance across all benchmark models and test datasets. The detection performance is robust across other LLMs such as Llama 3 and out-of-sample datasets. Collectively, these results suggest that our framework combining both semantic and linguistic features to capture the differences between AI and human texts outperforms alternative approaches involving only one type of information.

6. Model interpretability

In this section, we examined the interpretability of the proposed *SemLinDetector* framework using Layer Integrated Gradients (LIG) implemented in the Python library Captum [12,13], as such a model interpretability analysis can help more intuitively understand the differences between AI-generated and human-written texts. The LIG is an attribution technique that measures the contribution of the neurons in the deep neural network to the models' prediction by integrating gradients along the path from the baseline samples to the input at a specific layer in the network. In this paper, we attribute the *SemLinDetector*'s predictions to both the text embedding layer in the Semantic Encoder and all 16 Textagon linguistic feature embedding layers in the Linguistic Encoder.

We used both HC3-Test and Students & Synthetic AI as the datasets to produce LIG attribution scores to show that our model: (1) learns from the samples with same distribution and (2) generalizes the derived pattern to texts with different topics and genres. In our case, a high attribution score indicates that a specific word strongly influences the model by contributing to the classification of the text as AI-generated, while a low or negative score suggests that the word has little or negative influence on classifying the text as AI-generated. By analyzing these attribution scores in the training corpus, we identified the most influential words in original texts or tags in the parallel linguistic representations for distinguishing between AI-generated and human-written content.

Because LIG attribution scores measure attributions for AI-generated text decisions, we identified the highest scores for AI-generated texts and the lowest scores for human-written texts. Table 5 shows the top 10 words in AI-generated and human-written texts, respectively. In addition, we present a more comprehensive table featuring the top 30 words in Table A2 in Appendix A (additional features are available upon request). Our semantic pattern analysis below is based on the top 100-words. Tables 6 and 7 further present the top linguistic tags with the top LIG attribution scores in the categories of part-of-speech (POS) and named entity recognition (NER). By analyzing these most influential words and linguistic tags for AI-generated and human-written content, we identified some potential patterns to better understand the content generated by AI.

Regarding semantic patterns, AI-generated texts tend to include more connective or transition phrases that structure ideas logically. Tokens like “therefore,” “moreover,” “overall,” and “furthermore” suggest that AI writing tends to favor clear segmentation, possibly due to its training on structured texts such as articles and reports. In addition, AI-generated texts are more likely to use neutral and positive terms. Additionally, AI-generated texts show a preference for longer, more complex vocabulary over simpler synonyms. For instance, AI might opt for “evenly” instead of “finally” or “upon” instead of “on,” which may contribute to the polished yet slightly unnatural tone often associated with AI writing. Lastly, AI-generated texts are more likely to include abstract and formal words. Terms like “sync,” “integral,” “senator,” and “versatile” suggest an inclination toward abstract and general concepts, possibly due to the AI’s broad training across diverse topics.

In contrast, human-written texts are more likely to include colloquial words. The presence of tokens such as “did,” “say,” “always,” and “have” points to a more colloquial, informal style of writing, which is more typical of spontaneous human expression. Moreover, human-written texts are more indicative of personal emotion, like “hi,” “defeated,” “fights,” and “trying,” which is a typical characteristic of human writing in online forums. In addition, while AI tends to use more abstract words, human prefers to use specific words existing in the physical world, such as “villages,” “horses,” “cities,” and “building.” Lastly, human-written texts tend to use more words related to people such as “I,” “me,”

“you”, and “girls.” The reason might be the social function of human language and authors’ tendencies to position themselves relative to others.

Regarding parts of speech, compared to AI, human authors prefer to use auxiliary verbs (AUX) and adverbs (ADV) to convey nuanced tense, mood, and emphasis, while coordinating conjunctions (CCONJ) and pronouns (PRON) facilitate cohesion and personal expression. In contrast, AI-generated texts are often more formal, structured, and information-dense, favoring proper nouns (PROPN), determiners (DET), and symbols (SYM) to reference specific entities and maintain grammatical clarity. These differences arise from AI’s training on large-scale corpora with a focus on clarity and coherence, while human writing reflects personal intention.

The attribution of the NER tags reveals a similar pattern. The AI-generated texts prefer formal and abstract words like geopolitical entities (GPE), time, and measurable quantities (QUANTITY), while human authors prefer to use specific terms related to their daily life like money, person, facilities, or buildings (FAC).

The patterns derived from the LIG attribution to the textual words and Textagon linguistic features are consistent in both the HC3-Test from the online forum QA corpus and the student essays. Table 8 shows several examples of the AI and Human written texts from HC3-Test. The pattern is clear and consistent with what we have just discussed. The AI written texts use a clearer and more formal structure of sentences, while the human answers, like in the first example, use a more flexible structure. Also, humans tend to position themselves in social relationships and use more pronouns like “I” and “You.” In contrast, the AI-written texts focus more on the focal object itself, as shown in the two examples in Table 8.

7. Discussions and conclusions

In this paper, we proposed a novel detection framework, SemLin-Detector, which integrates a transformer-based semantic encoder (DeBERTaV3) and a linguistic encoder with parallel linguistic representations into a unified framework. By comparing with a set of benchmark models on datasets from various LLMs, our proposed model consistently demonstrates the most robust detection performance not only on the texts generated by a wide range of LLMs but also on a series of out-of-sample datasets in different areas. Further, interpretation of our proposed model via Layer Integrated Gradient helps better understand the reasoning behind the detector’s prediction outcomes, allowing individuals to make more reliable final decisions when evaluating online content via the detector.

7.1. Implications for research

Our study provides several important implications for related IS research in the areas of GenAI. First, we propose a novel detection model

Table 5
Top word-level LIG attribution scores.

AI-generated Text				Human-written Text			
HC3-Test		Students & Student Chosen AI		HC3-Test		Students & Student Chosen AI	
Word	Attr	Word	Attr	Word	Attr	Word	Attr
<i>evenly</i>	0.108	<i>conclusion</i>	0.057	<i>defeated</i>	−0.130	<i>aren</i>	−0.048
<i>aids</i>	0.080	<i>profound</i>	0.047	<i>hi</i>	−0.124	<i>was</i>	−0.044
<i>sync</i>	0.065	<i>upon</i>	0.046	<i>paradox</i>	−0.124	<i>compete</i>	−0.043
<i>senator</i>	0.061	<i>imagine</i>	0.043	<i>burned</i>	−0.112	<i>engagement</i>	−0.039
<i>sting</i>	0.061	<i>relatively</i>	0.035	<i>'s</i>	−0.098	<i>being</i>	−0.038
<i>reacts</i>	0.059	<i>industries</i>	0.035	<i>persons</i>	−0.084	<i>transmit</i>	−0.037
<i>personalized</i>	0.057	<i>integral</i>	0.032	<i>lands</i>	−0.081	<i>more</i>	−0.035
<i>promptly</i>	0.056	<i>instance</i>	0.031	<i>pointing</i>	−0.079	<i>them</i>	−0.034
<i>bears</i>	0.055	<i>thereby</i>	0.030	<i>leg</i>	−0.074	<i>were</i>	−0.034
<i>imagine</i>	0.055	<i>imperative</i>	0.030	<i>scandal</i>	−0.071	<i>me</i>	−0.032

Note: For better readability, we convert the tokens from the DeBERTaV3 tokenizer into full words for display in this table.

Table 6

Top POS tags LIG attribution scores.

AI-generated Text				Human-written Text			
HC3-Test		Students & Student Chosen AI		HC3-Test		Students & Student Chosen AI	
Word	Attr	Word	Attr	Word	Attr	Word	Attr
<i>propn</i>	0.054	<i>propn</i>	0.048	<i>aux</i>	−0.060	<i>aux</i>	−0.065
<i>det</i>	0.035	<i>det</i>	0.033	<i>adv</i>	−0.047	<i>cconj</i>	−0.064
<i>sym</i>	0.004	<i>sym</i>	0.003	<i>cconj</i>	−0.041	<i>adv</i>	−0.050
<i>verb</i>	0.002	<i>adp</i>	0.000	<i>pron</i>	−0.032	<i>pron</i>	−0.043
<i>adj</i>	0.000	<i>num</i>	0.000	<i>punct</i>	−0.018	<i>punct</i>	−0.025
<i>num</i>	−0.001	<i>verb</i>	−0.001	<i>noun</i>	−0.014	<i>sconj</i>	−0.022
<i>adp</i>	−0.003	<i>adj</i>	−0.003	<i>sconj</i>	−0.014	<i>noun</i>	−0.018
<i>intj</i>	−0.006	<i>intj</i>	−0.006	<i>part</i>	−0.011	<i>part</i>	−0.013
<i>part</i>	−0.011	<i>part</i>	−0.013	<i>intj</i>	−0.006	<i>intj</i>	−0.006
<i>sconj</i>	−0.014	<i>noun</i>	−0.018	<i>imperative</i>	−0.003	<i>adj</i>	−0.003

Table 7

Top NER tags LIG attribution scores.

AI-generated Text				Human-written Text			
HC3-Test		Students & Student Chosen AI		HC3-Test		Students & Student Chosen AI	
Word	Attr	Word	Attr	Word	Attr	Word	Attr
<i>gpe</i>	0.010	<i>time</i>	0.002	<i>ordinal</i>	−0.014	<i>money</i>	−0.011
<i>norp</i>	0.008	<i>event</i>	0.002	<i>fac</i>	−0.009	<i>ordinal</i>	−0.010
<i>work_of_art</i>	0.007	<i>gpe</i>	0.002	<i>money</i>	−0.008	<i>person</i>	−0.004
<i>time</i>	0.007	<i>norp</i>	0.002	<i>person</i>	−0.004	<i>product</i>	−0.003
<i>quantity</i>	0.006	<i>quantity</i>	0.002	<i>product</i>	−0.003	<i>org</i>	0.000
<i>event</i>	0.006	<i>work_of_art</i>	0.002	<i>org</i>	0.000	<i>cardinal</i>	0.000
<i>date</i>	0.005	<i>date</i>	0.002	<i>cardinal</i>	0.000	<i>law</i>	0.002
<i>loc</i>	0.004	<i>loc</i>	0.002	<i>law</i>	0.001	<i>loc</i>	0.003
<i>percent</i>	0.003	<i>law</i>	0.002	<i>percent</i>	0.003	<i>date</i>	0.004
<i>law</i>	0.001	<i>cardinal</i>	0.000	<i>loc</i>	0.004	<i>work_of_art</i>	0.005

Table 8

Two example pairs of human- and AI-written texts.

1	Prompt	Why do cars only have defroster grids in the rear window? I am thinking it has something to do with what might happen if the windshield shattered, but it would be convenient a lot of the time. Please explain like I'm five.
	Human Written	The latest models do have defroster in the front. Ford for example has trademarked its heated windshield under the name "Quickclear". So, its coming. Check [this] (URL,0) for example. Plus, you do n't wanna look to / thru all those little wires.
	Machine Written	Cars have defroster grids on the rear window to melt ice and improve visibility. The front windshield uses air blowers because wires could obstruct your view, ensuring safety while driving.
	Prompt	Why do laptops slow down so much after 3–4 years? Assuming no malware, viruses, etc. Explain like I'm five.
2	Human Written	During that time you sort of install more programs, memory gets fragmented, basically you get wear and tear. I recommend just re - installing windows and wiping the drive and it will run like new.
	Machine Written	Over time, laptops slow because parts get tired and can't keep up with new programs. Just like old toys can wear out, laptops do too!

to address the potential societal challenges of GenAI. GenAI such as LLMs may create misinformation, engender ethical issues, and generate misleading content due to bias or hallucinations [61,38,62]. Although prior research has developed a series of detection methods to effectively identify AI-generated texts, most of them mainly focused on one type of information rather than combining multiple types of information in the detection framework. Through the theoretical discussions of semantic and linguistic differences in AI vs. human writing, we argue that introducing the linguistic representations into a transformer-based model specialized in semantics could improve detection performance and enhance our understanding of AI-generated content. By combining both

semantic information and linguistic features into an end-to-end framework, we propose SemLinDetector, which achieves superior performance through theoretically extended design. The results suggest the importance of considering multiple types of information, especially linguistic features, in a detection framework for recognizing AI-generated content.

In addition, our proposed detection framework is characterized by cross-model robustness and enhanced model interpretability. GenAI is a continuously developing space, and LLMs can have variations across models and over time [63]. Therefore, it is important to design detection methods that can demonstrate consistent and reliable performance across different LLMs. Our experiments show that detection models trained on one LLM can be generally capable of detecting texts generated by other variations of LLMs, and this is particularly the case with our proposed framework which incorporates linguistic representations. Further, detection methods are subject to the risks of generating irresponsible predictions (i.e., black-box detection) [8,9]. Those who adopt a detection system without knowing the decision rationales of the detector may experience the loss of trust and a lack of confidence in the decision. Our proposed framework, in this regard, provides improved explainability in classifying AI vs. human content. These implications can further inform the design of responsible decision support systems to perform generalizable and reasonable AIGC detections.

Lastly, our model interpretability analysis uncovers several underlying characteristics of AI-generated texts. As an emerging new technology, GenAI system behaves differently from humans in terms of content generation, but little is known systematically regarding these differences. Our detection framework not only effectively identifies AI-generated texts but also helps to understand the underlying theoretical differences between AI-generated and human-written content. Our interpretability analysis emphasizes certain most salient patterns in AI-generated texts, offering insights into related literature to better understand the nature of GenAI.

7.2. Implications for practice

Our study also offers implications for practitioners in terms of regulating GenAI and addressing its risks. First, we develop a decision support tool for stakeholders operating in various fields such as media, digital platforms, education, and healthcare. This tool enables practitioners to systematically assess and regulate the use of GenAI-generated content. Our detection model shows improved performance over other benchmark models in multiple domains, and its performance is robust to variations of LLMs, helping content moderators and field practitioners continuously screen content without requiring constant adaptation. Furthermore, our detection model offers interpretability behind the detection decision through LIG. Content moderators can make more responsible decisions using the detection system and provide reasons to label content as AI-generated. Lastly, our insights about the patterns of AI-generated content help those without sufficient knowledge in screening AI content and understanding the detector system's decision rationales.

7.3. Limitations and future directions

Our study has several limitations that may be addressed in future research. First, our training dataset was constructed from online posts on social media and QA forums. Although our detection framework demonstrated strong performance on out-of-sample datasets, future research should incorporate more datasets in other contexts to enhance comprehensiveness. Second, our detection model mainly focuses on the raw content generated by AI. However, our model is subject to paraphrasing attacks, which involve making edits to the AI-generated content so that a detection model may not be capable of detecting the paraphrased AI-generated texts after revisions by a human or AI paraphrasing tool [8]. Third, our detection model is based on the responses from generic prompts collected from online postings. Prompt attacks can be a threat to our model because users may intentionally add specific instructions into the prompts to alter the semantic or linguistic patterns in responses. Future research may explore this challenge by developing detection methods robust to such prompt attacks.

Author note

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CRediT authorship contribution statement

Sen Yan: Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Zhiyi Wang:** Writing – review & editing, Supervision, Project administration, Formal analysis, Conceptualization. **David Dobolyi:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.dss.2025.114498>.

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Data availability

Data will be made available on request.

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